B.TECH PROJECT REPORT

GlandersWatch

Risk prediction, cited explanation and owner alerting for a neglected zoonosis

A report submitted in partial fulfilment of the requirements for the award of the Bachelor of Technology in Computer Science and Engineering

UNDER THE SUPERVISION OF

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Risk engine

a calibrated probability

Explanation layer

cited, or refused

Alert service

a message still worth reading

Department of Computer Science and Engineering · November 2026

A disease that still has no forecast

01

The organism

Glanders is a bacterial zoonosis of equids caused by Burkholderia mallei. Notifiable, frequently fatal, and with no environmental reservoir: it persists in animal populations rather than in soil or water.

02

The gap

Surveillance describes what has already happened. The published machine-learning work on this disease stops at laboratory identification. No population-level risk model has been reported.

Indian surveillance, 2019 to 2023, found 728 seropositive equids in 138 districts, sharply concentrated geographically. Cases cluster, and that is what makes a spatial model worth building.

What this project adds

A calibrated risk model, an explanation layer grounded in official guidance, and an alert service, each independently deployable behind a fixed contract.

Each component communicates through a versioned schema rather than shared internals, so none of the three hardcodes the disease and each degrades without failing the others.

WHY IT MATTERS

Population at risk is not the same as damage

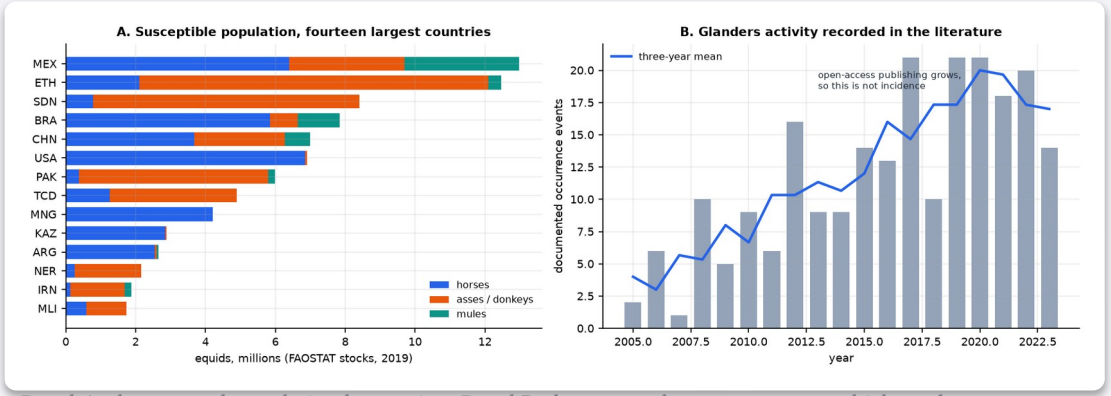
5.50 lakh
equids in India, 2019

728
seropositive, 2019–23

138
districts affected

The two figures are routinely conflated, so they are separated here at the outset.

- 5.50 lakh is the exposed population: how many animals are susceptible.
- 728 of 158,798 screened is the documented damage: under one per cent.
- Each positive is acted on rather than monitored, because the disease is notifiable and often fatal once clinical.



Panel A: the exposed population by species. Panel B: documented events per year, which tracks open-access publishing as much as disease.

Because documentation follows publishing capacity, the target is a documented event rather than raw incidence. That choice is stated once and held throughout.

DEFINITION

Problem statement

Given a country's features in year t, predict whether glanders is documented in that country in year t + 1.

01

The label

One country-year is positive if a glanders occurrence is reported from that country in that year. Features come only from the year before, so the model never reads the outcome it is predicting.

02

The difficulty

169 positives in 2,880 rows. A model that always says “no” is 94 per cent accurate and worthless, which is why accuracy and ROC-AUC are not used to rank models here.

Base rate 5.87 per cent. Precision-recall AUC is the primary metric because it moves when a model starts finding the 169 positives; ROC-AUC does not.

Every later result is reported on this target and this metric. The country-level model in the next section asks an easier question and is labelled as such.

SCOPE

Project objectives

01

Assemble the panel

Four public sources, none previously applied to this disease, into a country-year panel.

02

Build the risk engine

Select features by attribution, compare six model families under two splits, calibrate.

03

Specify the explanation layer

Retrieval grounded in official guidance, with citations and a refusal protocol.

04

Specify the alert service

Levels from the calibration curve, hysteresis, deduplication, delivery logging.

Objectives 01 and 02 carry measured results. 03 and 04 are contracts with an evaluation protocol, and are never presented as measured.

USE

Who the system is for

01

District veterinary officer

Where to pre-position testing and movement checks, before the report arrives.

02

Equine owner

Whether this is the month to restrict movement, in a language they read.

03

State planner

A single scale on which districts can be compared for resources.

The failure mode to design against

A warning system that fires for every area every month is ignored by month twelve. Alert budget is therefore a design constraint, not a tuning parameter.

None of the three users operates a dashboard. The output has to arrive as a message, and the reasoning has to arrive with it.

01

The risk engine

Can a country's next year be forecast from its last?

- Four public sources assembled into a country-year panel
- Eight features chosen by attribution rather than correlation
- Six model families, two validation splits, and a calibration check

DATA

Four public sources, none used for this disease

01

Europe PMC

Occurrence events extracted from abstracts. This is the label.

02

FAOSTAT

Horse, ass and mule head counts. This is exposure.

03

World Bank

Development and connectivity indicators. This is context.

04

NASA POWER

Temperature, humidity and rainfall. This is environment.

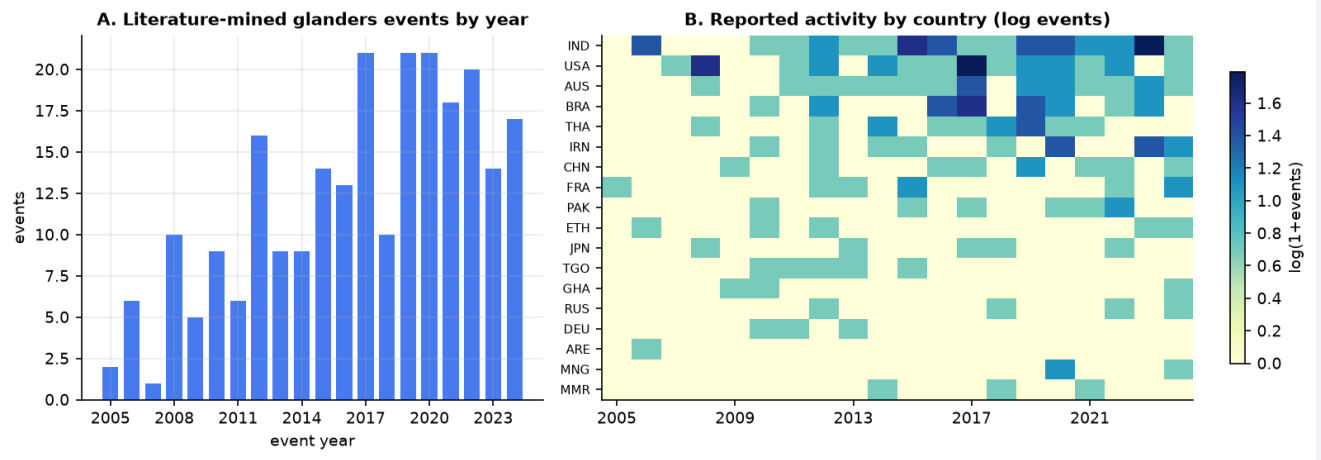
2,880
modelling rows

160
countries

169
positives

5.87%
prevalence

6,700 raw country-year rows over 268 reporting entities reduce to 2,880 observations. Rows with no climate baseline are dropped rather than imputed, so no anomaly is invented.



Coverage of the assembled panel: which countries contribute which years.

METHOD

From raw sources to a calibrated probability

01 Sources
Four APIs, one panel

02 Panel
Country-year rows

03 Attribution
42 ranked, 8 kept

04 Training
Six families, two splits

05 Output
Calibrated probability

The temporal split trains on 2006–2017 and tests on 2018–2023: forward in time, which is the only honest test of a forecast.

The country split holds out whole countries, so no row from a test country appears in training. This is the split that matches deployment.

Feature ranking uses training years only. Fitting on the full panel would let the held-out years choose the features and make the temporal score optimistic.

Model family	Character
Logistic Regression	additive, interpretable
Decision Tree	single split tree
Random Forest	bagged trees
Extra Trees	randomised splits
XGBoost	gradient boosted
LightGBM	leaf-wise boosted

Eight numeric features, no categorical columns, no identifiers. Country codes and calendar years are identifiers and never enter the model.

FEATURES

Eight features, chosen by attribution

#	Feature	Unit as measured	Mean |SHAP|
1	pop_total	persons	1.035
2	years_since_last	years (99 = never)	0.648
3	temp_mean_anom	℃ from baseline	0.505
4	internet_pct	% of population	0.449
5	humidity_anom	% RH from baseline	0.382
6	precip_cv	ratio, sd / mean	0.371
7	log_livestock	ln(1 + LSU)	0.364
8	goats	head	0.362

8 of 42 numeric candidates, carrying 49 per cent of total attribution.

Population and years since the last report rank first: exposure, then history.

Attributes a prediction

SHAP says how far each feature moved this model's output, so the ranking matches what the model uses.

Beyond pairs

Correlation and mutual information score features pairwise; they cannot see interactions or redundancy.

Comparable

Both are unnormalised and undirected. Attribution is additive and sums to the prediction.

Correlation ranks longitude first and attribution ranks it near the bottom. Selecting on association keeps geography proxies instead of drivers.

MODELS

The three models worth describing

01

Logistic Regression

A weighted sum of the eight features passed through a sigmoid. The coefficients are readable, which matters when a prediction has to be defended.

$$p = 1 / (1 + e^{-z})$$
$$z = \beta_0 + \beta_1 x_1 + \ldots + \beta_8 x_8$$

02

Random Forest

Five hundred trees, each grown on a bootstrap sample with a random feature subset at every split, then averaged. Captures interactions, and cannot extrapolate beyond the data it saw.

$$\hat{y} = (1/B) \sum T_b(x)$$
$$G = 1 - \sum p_k^2$$

03

Gradient Boosting

Trees added in sequence, each fitting the residual of the last. XGBoost and LightGBM differ mainly in how they grow the trees.

$$Obj = \sum L(y_i, \hat{y}_i) + \sum \Omega(f_k)$$
$$\Omega(f) = \gamma T + \tfrac{1}{2}\lambda \|w\|^2$$

Six families are trained in total; Decision Tree and Extra Trees are reported in the results table but add nothing to the argument. Logistic Regression and Random Forest are the two that carry it: one is additive and stable, the other flexible and unstable across countries.

Random Forest wins on unseen years; Logistic Regression wins on unseen countries. The next slide is about why that inversion is the finding.

RESULTS

One target, two validation splits

Model	PR-AUC, unseen years	PR-AUC, unseen countries
Logistic Regression	0.461	0.381
Decision Tree	0.376	0.243
Random Forest	0.495	0.322
Extra Trees	0.494	0.353
XGBoost	0.429	0.268
LightGBM	0.471	0.280

Base rate 5.87 per cent, so PR-AUC is the metric that moves.

ROC-AUC stays inside 0.80–0.87 for all six families on both splits. It cannot separate them, which is why it is not the headline.

Logistic Regression is the champion because it wins on unseen countries, which is what deployment means.

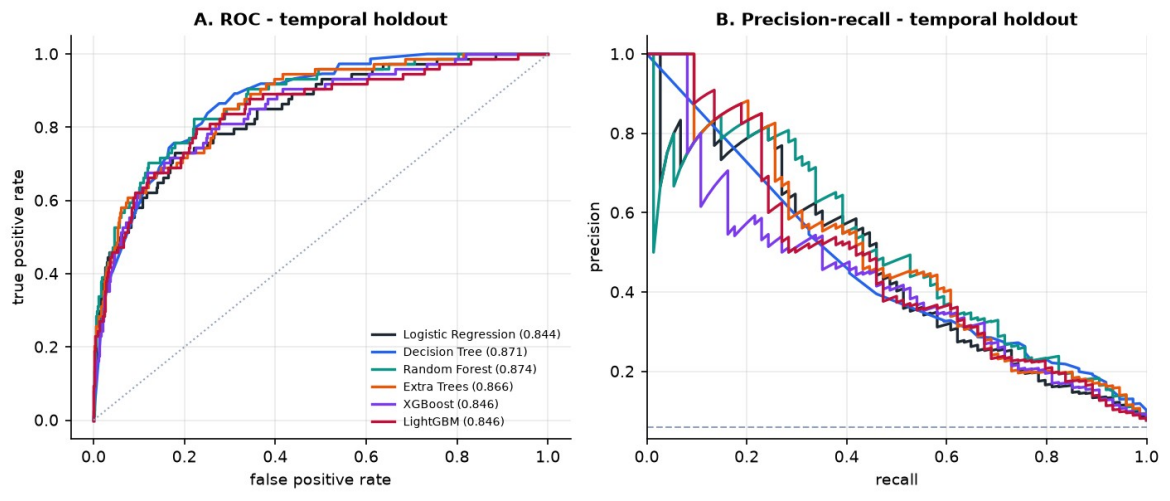
0.495
random forest, unseen years

0.381
logistic, unseen countries

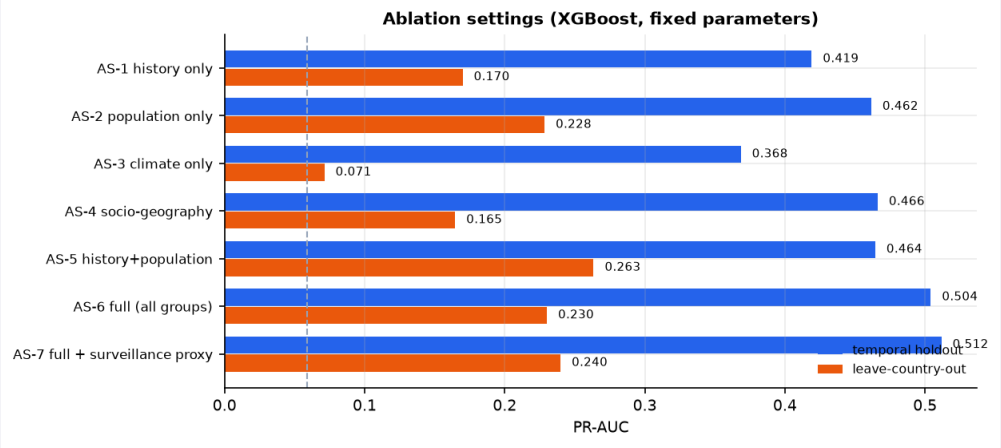
0.322
random forest, unseen countries

The ranking inverts between the two splits. A model that generalises across countries has to be simpler than one that generalises across years, because country-level structure is what the flexible models memorise.

What the two metrics actually measure



Left: ROC curves, which stay high because the negatives vastly outnumber the positives. Right: precision-recall, which falls as recall rises.



Attribution-based selection against correlation-based, on the same split.

The ROC panel is why the deck does not lead with ROC-AUC. Every family sits near 0.85 there, including models that find almost none of the positives. The PR panel separates them, and the ablation shows what the feature choice is worth.

Two metrics, two different questions, and only one of them tells you whether the model finds events.

CALIBRATION

The number has to be worth quoting

0.025
expected calibration error

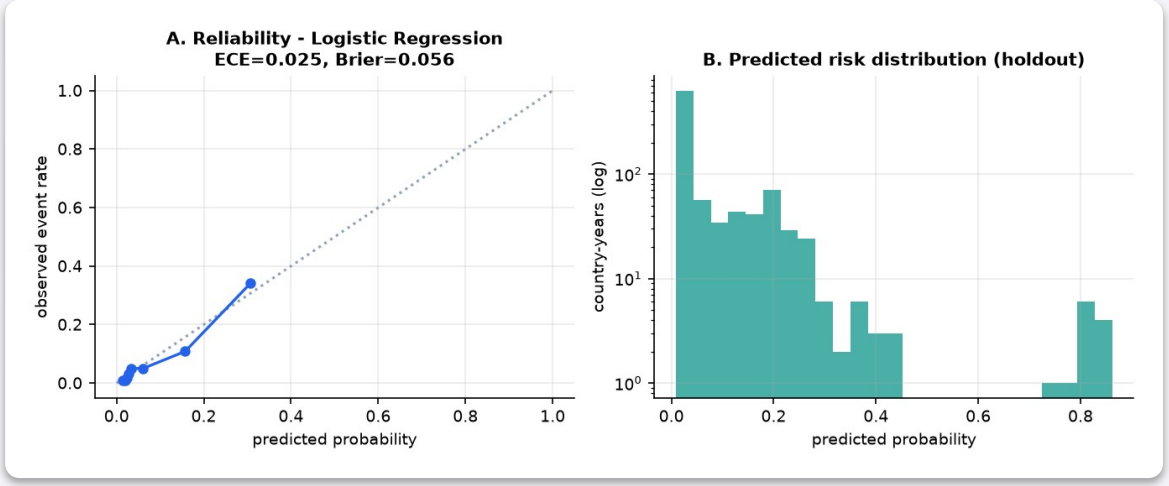
0.056
Brier score

43%
events caught at a 6% budget

A ranking can sort countries. A message needs a number somebody can act on, and that requires the probabilities to mean what they say.

An error of 0.025 means a predicted 0.20 corresponds to roughly one event in five. The reliability curve below is the evidence.

The alert levels are read off this curve rather than set to a default 0.5.



Predicted probability against observed frequency. A well calibrated model follows the diagonal.

Calibration is what lets one number serve two consumers: the alert service thresholds it, and the explanation layer quotes it back to a user.

SUPPLEMENTARY

An easier question, answered better

Model	ROC-AUC	PR-AUC
Logistic regression	0.951	0.914
Random forest	0.968	0.963

161 countries, 56 ever positive, prevalence 34.8 per cent.

Discrimination rises to about 0.97, and that is not a better model. It is a simpler question: whether a country reports the disease at all is a stable national property.

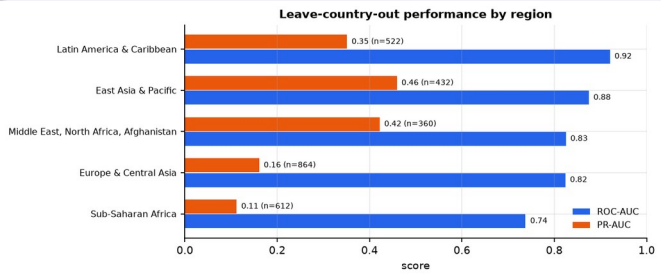
Asking for the year instead

Predicting the calendar year of a report scores $R^2 = -0.103$ under a linear model.

Target	Model	R^2
Year of report	Linear	−0.103
Year of report	Random forest	−0.095
Year of report	Predict the mean	0.000

Why it fails

Negative R^2 means worse than predicting the average. The year tracks when reporting began, not biology.



The negative result is kept in the report because deleting it would hide how much of the signal is reporting practice rather than disease.

02

The explanation layer

Why did the model flag this country, and what should an owner actually do?

- Grounded retrieval with citations, and refusal when nothing supports an answer
- Guardrails that stop advisory text from inventing guidance
- The evaluation protocol, and what has not yet been measured

EXPLANATION

An answer is retrieved, then written

01

Question

"Why is this district high risk?"

02

Retrieve

Dense plus lexical search over the guidance corpus

03

Fuse

Reciprocal rank fusion, then a cross-encoder rerank

04

Answer

Every claim tied to a retrieved passage, or refused

Guardrails, not suggestions

Refuses to answer outside the retrieved sources. Refuses to diagnose an individual animal. Always shows the passages it used.

Why retrieval rather than a fine-tune

Guidance changes and weights would have to be retrained to follow it. A cited passage can be checked by a veterinarian; a generated sentence cannot.

"Risk is elevated mainly because of recent reported activity in neighbouring districts and high equine density. Restrict equid movement and contact the local veterinary authority if an animal is suspected."

EXPLANATION

How this layer will be judged

Metric	What it asks	How it is measured
Recall@k	Is the right passage in the top k?	Hand-built question set
Faithfulness	Does the answer follow from those passages?	RAGAS, plus manual review
Answer relevance	Does it answer the question asked?	RAGAS
Citation precision	Does every claim trace to a passage?	Automated check

These four cover the ways an explanation fails: retrieving the wrong passage, drifting from the passages it retrieved, or faithfully answering a question nobody asked.

None of these are measured yet. Targets are fixed before measurement, so the layer cannot be scored by choosing a threshold after seeing the results.

Why this is stated rather than shown

The layer is specified to the level of its interface and rules. Reporting a metric without a deployment would be a stronger claim than the evidence supports.

03

The alert service

How does a probability become a message that is still read in month twelve?

- Levels read from the calibration curve, not a default cutoff
- Hysteresis, deduplication and a stale-data guard
- Delivery measured against alert fatigue, not accuracy

ALERTING

Five stages, each preventing one failure

01 — Predict
Latest calibrated score for the area

02 — Assign level
Thresholds from the budget curve

03 — Deduplicate
Three idempotency keys

04 — Deliver
SMS or WhatsApp, with retries

05 — Log
Status per recipient

Hysteresis

Enter at a threshold, leave only 0.05 below it, so a country on the line does not oscillate.

Freshness

If the data check fails, no alert goes out and the admin is told instead.

Quiet hours

Hold overnight except for escalation, so a real rise still gets through.

9%
budget for half the events

37%
budget for 80 per cent

43%
caught at a 6 per cent budget

Levels carry a measured meaning because they come from the curve, and deduplication is what stops a re-run or a retrain from sending the same warning twice.

What an owner receives, and how it is judged

Metric	What it asks
Precision	Of the alerts sent, how many were real events?
Recall	Of the real events, how many did we alert on?
Missed events	The failure a user actually feels
False alerts per subscriber	The alert-fatigue budget

Defined but not yet measured. A pilot with a district office is how these acquire numbers, and until then the service is not claimed to work.

GLANDERS RISK ALERT

Risk
HIGH

Reason
recent reported activity in neighbouring districts; high equine density

Action
avoid unnecessary movement or mixing of equids; contact the local veterinary authority if an animal is suspected of illness

Caveat
a population-level forecast, not a diagnosis

The message names its reason and its action, and says plainly that it is a forecast. A warning without those three parts is not actionable, and an owner who cannot act on it will stop reading.

CONCLUSION

What is built, and what a pilot would settle

Conclusion

The risk engine is built and measured: six families, two splits, calibrated, with every number reproducible from committed scripts. The explanation and alert layers are specified to the level of their interfaces and rules.

Future scope

Model reporting capacity as a quantity rather than a covariate. Measure the explanation layer on a hand-built question set. Pilot the alert service with a district office and report precision, recall and fatigue.

The dominant limitation is the label: a negative may mean no disease or no report. With 169 positives the uncertainty on every headline metric is wide, and the honest reading is that this is a moderate model on a hard, noisy target.

The model is the part that is finished. Evidence that the explanation and alert layers behave in use is the part that is not, and a pilot is how it arrives.

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